

AI/ML Interview Questions

Regression Interview Questions

1.What is regression in machine learning, and when is it used?

A.Regression is supervised machine learning technique which is used to predict the Continuous value.It is used when the targeting column is continuous.

2.What is the difference between simple linear regression and multiple linear regression?

A.Simple linear regression is used when one independent variable and one dependent variable.

Multiple linear regression is used when multiple independent variables are there.

3.What assumptions are made by linear regression?

A.Linear regression assumes linearity between variables, independence of errors, homoscedasticity (constant variance), and normally distributed errors. It also assumes no multicollinearity among predictors.

4.What is the difference between a dependent variable and an independent variable?

A.The **dependent variable** is the output you want to predict. The **independent variable(s)** are the inputs/features used to make that prediction.

5.What do the slope and intercept represent in a linear regression model?

A.The **slope** shows how much the dependent variable changes for a unit change in the independent variable. The **intercept** is the value of the dependent variable when all inputs are zero.

6.What is the purpose of the cost function in regression?

A.The cost function measures how well the model fits the data. It quantifies the error between predicted and actual values and is minimized during training.

7.What is Mean Squared Error, and why is it commonly used for regression?

A.MSE is the average of squared differences between predicted and actual values. It is widely used because it penalizes large errors more strongly.

8.Difference between MAE, MSE, and RMSE?

A.**MAE** measures average absolute error (robust to outliers).

MSE squares errors (penalizes large errors more).

RMSE is the square root of MSE, giving error in original units.

9.What is R-squared?

A. R^2 measures how much variance in the dependent variable is explained by the model. Values range from 0 to 1, with higher values indicating better fit.

10.What is adjusted R-squared?

A. R^2 measures how much variance in the dependent variable is explained by the model. Values range from 0 to 1, with higher values indicating better fit

11.What is overfitting?

A.Overfitting occurs when a model learns noise and details from training data, performing well on training but poorly on new data.

12.What is underfitting?

A.Underfitting happens when a model is too simple to capture the data patterns, leading to poor performance on both training and test data.

13. What is multicollinearity?

A.Multicollinearity occurs when independent variables are highly correlated. It makes coefficient estimates unstable and hard to interpret.

14.How can multicollinearity be detected?

A.It can be detected using **correlation matrices**, **Variance Inflation Factor (VIF)**, or observing unstable regression coefficients.

15.What is regularization?

A.Regularization adds a penalty to the cost function to prevent overfitting. It helps keep model weights small and improves generalization.

16.Difference between Ridge, Lasso, Elastic Net?

A.**Ridge (L2)** shrinks coefficients but keeps all features.

Lasso (L1) can shrink some coefficients to zero (feature selection).

Elastic Net combines both L1 and L2 penalties.

17. How does Lasso perform feature selection?

A.Lasso adds an L1 penalty that forces some coefficients to become exactly zero. Features with zero coefficients are effectively removed.

18.What is polynomial regression?

A.Polynomial regression models non-linear relationships by adding polynomial terms (e.g., x^2 , x^3) while still using linear regression techniques.

19.How do outliers affect regression?

A.Outliers can heavily influence the model, especially in MSE-based methods. They can skew the regression line and reduce model accuracy.

20.How would you evaluate a regression model?

A.Use metrics like **MAE**, **MSE**, **RMSE**, **R^2** , and cross-validation.